Exercises: Design, aggregation and safe code

Exercises 1 and 2 are written answers, and so is the last one. The rest are done in Python. Put the header below at the top of every file. Start by running python seed_webshop.py so everyone has the same data. Look back at the slides when you need the syntax.

The header (at the top of every Python file):

```
from pymongo import MongoClient

client = MongoClient("mongodb://localhost:27017/")
db = client["webshop"]
```

Exercise 1

Design: a library app shows a book page with the book's title, author and all its reviews. Many members borrow the same book. Would you embed or reference the reviews? The members? Explain in two sentences.

Expected: written answer

Exercise 2

Our orders copy the product name into each item. Give one advantage and one disadvantage of that copy.

Expected: written answer

Exercise 3

Create a collection newsletter_signups with validation: email is required and must be a string, and source can only be "web" or "store" (look up enum). Insert one valid and one invalid document.

Expected: the second insert fails

Exercise 4

How many customers live in each city? Use $group. Biggest city first.

Expected: 6 rows (one with _id None)

Exercise 5

What is the cheapest and the most expensive product in each category? Use $min and $max.

Expected: 3 rows

Exercise 6

How many orders does each customer have (all statuses)? Show only customers with 2 or more orders. Hint: $match after $group works like HAVING.

Expected: 5 rows

Exercise 7

For each order, show the _id and the number of items in it. Use $project with $size.

Expected: 15 rows

Exercise 8

How many units of each category have been sold? Products' categories are not in the orders, so you need $unwind and $lookup to products. Skip cancelled orders.

Expected: 3 rows

Exercise 9

Revenue per status: for each status, the total of quantity times unit_price.

Expected: 4 rows

Exercise 10

Challenge: which tags are the most common? Show each tag and how many products have it, most common first.

Expected: 7 tags

Exercise 11

Webshop: this function is unsafe. Explain why, and fix it.

```
def find_orders(request_text):
    data = json.loads(request_text)
    return list(db.orders.find({"status": data["status"]}))
```

Test with '{"status": "new"}' and with '{"status": {"$ne": "x"}}'.

Expected: 4, then 0 (the unsafe version gives 15)

Exercise 12

Webshop: write a function products_under(text) for a search box. The text comes from the user, so convert it with float(). If the conversion fails, catch the ValueError, print 'Please enter a number.' and return an empty list. Otherwise return the name and price of all products cheaper than that price.
Test it with "300", "abc" and '{"$gt": 0}'.

Expected: 5 products for "300". For the other two: Please enter a number.

Exercise 13

What happens when the database is down? Make a second client that points to the wrong port: MongoClient("mongodb://localhost:27018/", serverSelectionTimeoutMS=2000). Count the products with it inside try/except, catch ServerSelectionTimeoutError (from pymongo.errors) and print a friendly message instead of a crash. Then do the same count with the normal client.

Expected: After about 2 seconds your friendly message. Then 12 with the normal client.

Exercise 14

Written answer: personal data in the webshop. Which fields in customers are personal data? Orders only store customer_id, not the customer's name or email. Why is that good when a customer asks to be deleted? A classmate wants to copy the customer's email and city into every order to make reading faster. What do you answer? Write 4-6 sentences.

Expected: A written answer. Compare with a classmate.